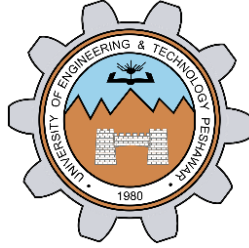


LAB # 09

Keypad interfacing (port multiplexing)



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CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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May 12th, 2024

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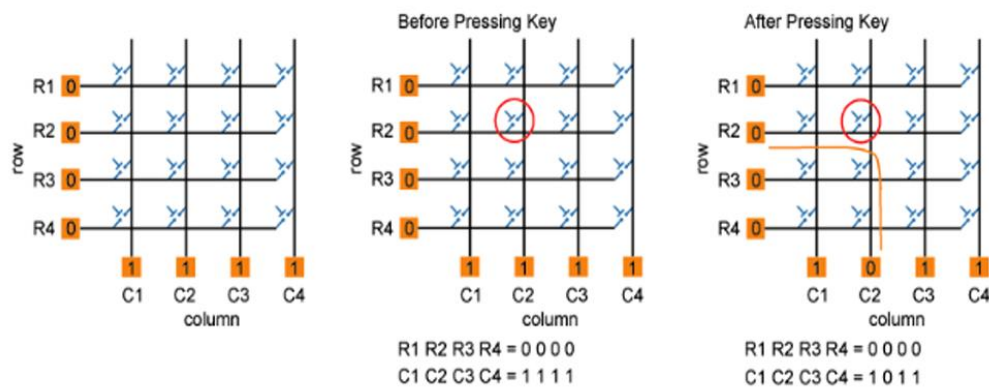
Objectives of The Lab:

- keypad interfacing with 8051 microcontroller

Keypad:

- Keyboards are organized in a matrix of rows and columns.
- When a key is pressed a row and a column make contact, otherwise there is no connection between them.
- With 8-bit PORT 4x4 matrix of keys can be connected to a microcontroller.

Keypad Matrix Basics



Working:

- The hex keypad has 8 communication lines namely R1, R2, R3, and R4
C1, C2, C3, and C4
- R1 to R4 represents the four rows.
- C1 to C4 represents the four columns.
- When a particular key is pressed the corresponding row and column to which the terminals of the key are connected gets shorted.
- For example, if key 1 is pressed row R1 and column C1 gets shorted and so on.

Key Scanning:

- To detect a pressed key, the microcontroller grounds all rows by providing 0 to the output latch, and then it reads the columns.
- If the data read from columns is = 1111, no key has been pressed
- After pressing key, it makes contact of row with column.
- If one of the column bits has a zero, this means that a key press has occurred. For example, if C1:C4 = 1011, this means that a key in the C2 column has been pressed.
- After detecting a key press, microcontroller will go through the process of identifying the key.

- Starting from the top row, the microcontroller will ground it by providing a low to row R1 only.
- Now read the columns, if the data read is all 1s, no key in that row is pressed and the process continues for the next row.
- So, now ground the next row, R2. Read the columns, check for any zero and this process continues until the row is identified. E.g. In above case we will get row 2 in which column is not equal to 1111. So, after identification of the row in which the key has been pressed, we can easily find out the key by row and column value.

Lab Total Tasks:

1) Display Numbers from 1 to 9 on the seven-segment display and ON the corresponding LED's attached with P1.

Note: Attach the seven segment with P3 and keypad with P1.

Task 1:

Display Numbers from 1 to 9 on the seven-segment display and ON the corresponding LED's attached with P1.

Note: Attach the seven segment with P3 and keypad with P1.

C Code:

```

1  #include<reg51.h>
2  sbit led0 = P1^0;
3  sbit led1 = P1^1;
4  sbit led2 = P1^2;
5  sbit led3 = P1^3;
6  sbit led4 = P1^4;
7  sbit led5 = P1^5;
8  sbit led6 = P1^6;
9  sbit led7 = P1^7;
10 sbit R0=P2^0;
11 sbit R1=P2^1;
12 sbit R2=P2^2;
13 sbit R3=P2^3;
14 sbit C0=P2^4;
15 sbit C1=P2^5;
16 sbit C2=P2^6;
17

```

```
19 void main(void) {  
20     while(1) {  
21         P1=0x00;  
22         R0=0;  
23         R1=1;  
24         R2=1;  
25         R3=1;  
26         if(R0==0) {  
27             if(C0==0) {  
28                 P3=0xF9;  
29                 led0=1;  
30             }  
31             if(C1==0) {  
32                 P3=0xA4;  
33                 led1=1;  
34             }  
35             if(C2==0) {  
36                 P3=0x30;  
37                 led2=1;  
38             }  
39         }  
    }
```

```
40         R0=1;  
41         R1=0;  
42         if (R1==0) {  
43             if (C0==0) {  
44                 P3=0x19;  
45                 led3=1;  
46             }  
47             if (C1==0) {  
48                 P3=0x12;  
49                 led4=1;  
50             }  
51             if (C2==0) {  
52                 P3=0x02;  
53                 led5=1;  
54             }  
55         }  
    }
```

```

56     R1=1;
57     R2=0;
58     R3=1;
59     R0=1;
60     if(C0==0 && C1==1 && C2==1){
61         P3=0x78;
62         led6=1;
63     }
64     if(C0==1 && C1==0 && C2==1){
65         P3=0x00;
66         led7=1;
67     }
68     if(C0==1 && C1==1 && C2==0){
69         P3=0x18;
70     }
71     R2=1;
72     R3=0;
73     if(C0==1 && C1==0 && C2==1){
74         P3=0x40;
75     }
76 }
77 }

```

Proteus Simulation:

